Cook County Sample — Shopping Center / Retail

Traffic Impact Study

Prepared August 28, 2026 · 42.0334, -88.0834

Simple Impact Studies — sample draft

Traffic Impact Study

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EXECUTIVE SUMMARY

This Traffic Impact Study (TIS) presents the anticipated traffic impacts of the proposed Cook County Sample — Shopping Center / Retail located within Chicago-Naperville-Elgin MSA, Illinois. The study evaluates 13 intersections within a 1.00-mile study radius using an openly-published signalized control-delay method (Webster 1958 uniform delay with an Akcelik overflow term) and the public-data screening rates (NHTS 2017 / SANDAG 2002 / NCHRP 716). Trip generation is calculated for land use 820 (Shopping Center / Retail) at a development size of 85 1,000 sqft GLA.

Reviewing authority: Cook County DOTH.

Cook County Department of Transportation & Highways (DOTH) — Permits Division, per the Construction Permit Packet (Nov 2020). Cook County publishes no standalone TIS manual; TIS scope is staff-discretionary during the access/signal permit review.

Findings:

• 0 intersections are projected to drop one or more LOS grades under build conditions.

• 11 intersections operate at LOS E or F under build conditions and may require mitigation per BLRS Ch. 32 + Ch. 34 and the host-jurisdiction standard above.

13	0	11	28.0s
INTERSECTIONS	LOS DROPS	AT LOS E/F	WORST Δ DELAY

1.0 INTRODUCTION

This Traffic Impact Study follows the IDOT District 8 High-Volume Access-Permit Guidelines (April 2024) Appendix A as the base section structure — the only IDOT-published prescriptive TIS content list located. The report is layered with the Cook County DOTH overlay where applicable. Cook County Department of Transportation & Highways (DOTH) — Permits Division, per the Construction Permit Packet (Nov 2020). Cook County publishes no standalone TIS manual; TIS scope is staff-discretionary during the access/signal permit review.

Trigger basis: Staff-discretionary during the access/signal permit review (no published numeric trigger).

Note: Illinois has no single statewide TIS manual. The methodology applied here is assembled from IDOT BLRS chapters (Ch. 17 planning, Ch. 27 design controls + LOS, Ch. 28 sight distance, Ch. 32 geometric tables, Ch. 34 intersections, Ch. 39 traffic-control devices, Ch. 41 driveways), Title 92 Illinois Admin. Code Part 550 (driveway permit policy), and the IDOT District 8 April 2024 guidelines. District 1 (Schaumburg) publishes NO TIS-specific guideline document — only the D1 Traffic Signal Design Guidelines (October 2025), which govern signal design rather than TIS scope; D8 Appx. A is the de facto fallback statewide. D1 may still impose unwritten variations on growth-rate convention, software choice at IDS phase, and timeline — confirm at the kickoff meeting with the District Permits Unit Chief.

2.0 PROJECT DESCRIPTION

Project name	Cook County Sample — Shopping Center / Retail
Land use	LU 820 — Shopping Center / Retail
Development size	85 1,000 sqft GLA
Site coordinates	42.0334°, -88.0834°
Opening year	2027
Design year (opening + 20)	2047
Region	Chicago-Naperville-Elgin MSA
Host jurisdiction	Cook County DOTH

3.0 EXISTING CONDITIONS

Existing roadway geometry, posted speed, functional classification, lane count, and historical AADT are referenced from the IDOT Getting Around Illinois public AADT viewer (gettingaroundillinois.com) and the IDOT AADT GIS open-data layer. Crash history (3-year minimum) is sourced from the IDOT Safety Data Mart (consultant access via FOIA). Existing peak-period turning-movement counts (TMCs) and 24-hr machine counts within the most recent 12 months should be collected per D8 Appx. A — three-to-four peak-period hours minimum, Tuesday/Wednesday/Thursday, clear-and-dry conditions.

Affected intersection	Distance (mi)	Existing LOS	Existing delay (s)
North Roselle Road & Bethel Lane	0.25	F	196.7
West Schaumburg Road & North Pleasant Drive	0.43	F	162.1
South Roselle Road & West Schaumburg Road	0.45	F	196.7
Branchwood Drive & West Schaumburg Road	0.50	F	162.1
Hilltop Drive & West Schaumburg Road	0.60	F	162.1
South Roselle Road & Illinois Avenue	0.65	F	205.5
North Roselle Road & Bode Road	0.68	F	196.7
Summit Drive & North Summit Drive	0.81	B	14.8
North Roselle Road & West Higgins Road	0.82	F	216.4
Ash Road & West Higgins Road	0.88	F	300.0
Grand Canyon Parkway & West Higgins Road	0.88	F	216.4
South Salem Drive & West Schaumburg Road	0.96	B	17.1
Near North Roselle Road	0.98	F	300.0

4.0 TRIP GENERATION

Trip generation is calculated per the public-data screening rates (NHTS 2017 / SANDAG 2002 / NCHRP 716) for land use 820 (Shopping Center / Retail) at a proposed size of 85 1,000 sqft GLA. IDOT District 8 Appendix A requires "the current edition of the ITE Trip Generation Manual" for a submitted study; the figures here are public-data screening estimates (NHTS 2017 / SANDAG 2002 / NCHRP 716) and must be re-run with the jurisdiction-required source before submittal. Pass-by and internal-capture credits are taken from standard screening methodology and applied only against the external trips assigned to the study network. Supplemental sources are allowed for land uses not represented in ITE, with District permission.

Period	Entering trips	Exiting trips
Daily	3,400	3,400
AM peak hour	166	106
PM peak hour	326	354

Pass-by capture applied	30%
Internal capture applied	0%

Period	Raw	Pass-by	Int. cap.	External	In	Out
AM Peak Hour	272	20	0	123	75	48
PM Peak Hour	680	204	0	233	112	121
Saturday Midday	748	56	0	338	169	169
Daily Total	6,800	510	0	3,073	1,537	1,536

The external / net-external column is net-new vehicle trips assigned to the roadway = (gross − pass-by − internal-capture) × auto-mode share (48.8% for this metro). Walk / bike / transit person-trips are excluded from off-site assignment, so In + Out equals the external column.

Trip Distribution by Time of Day

Estimated within-day distribution of the 6,800 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

Figure — Inbound / Outbound Trips by Start Time

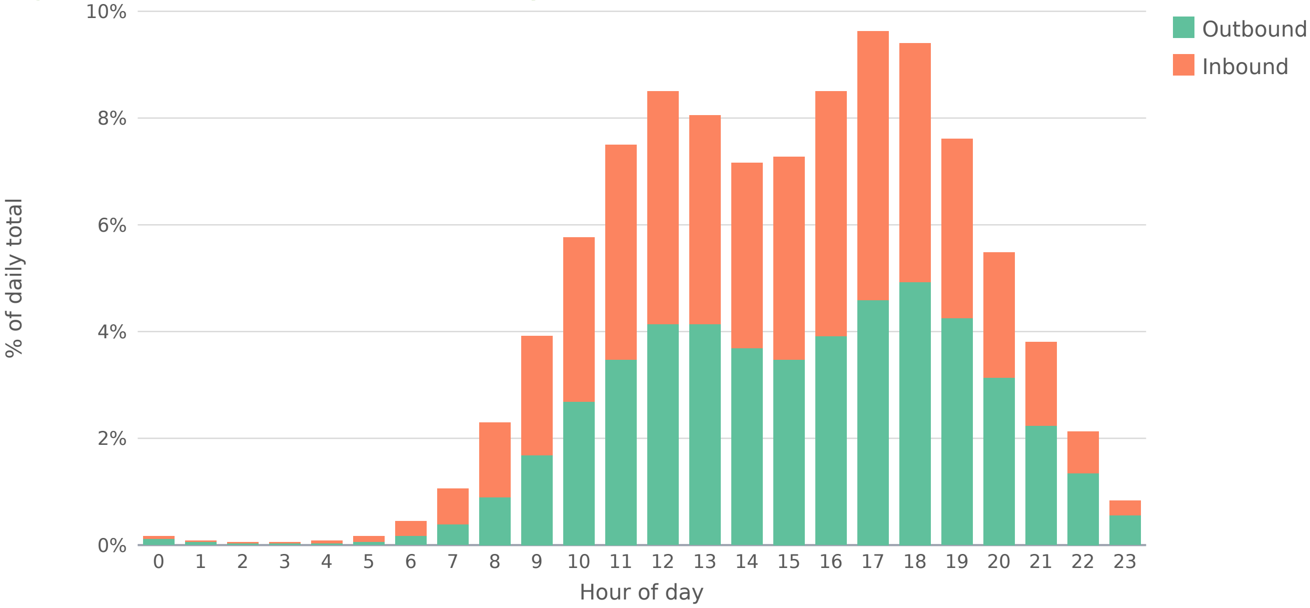
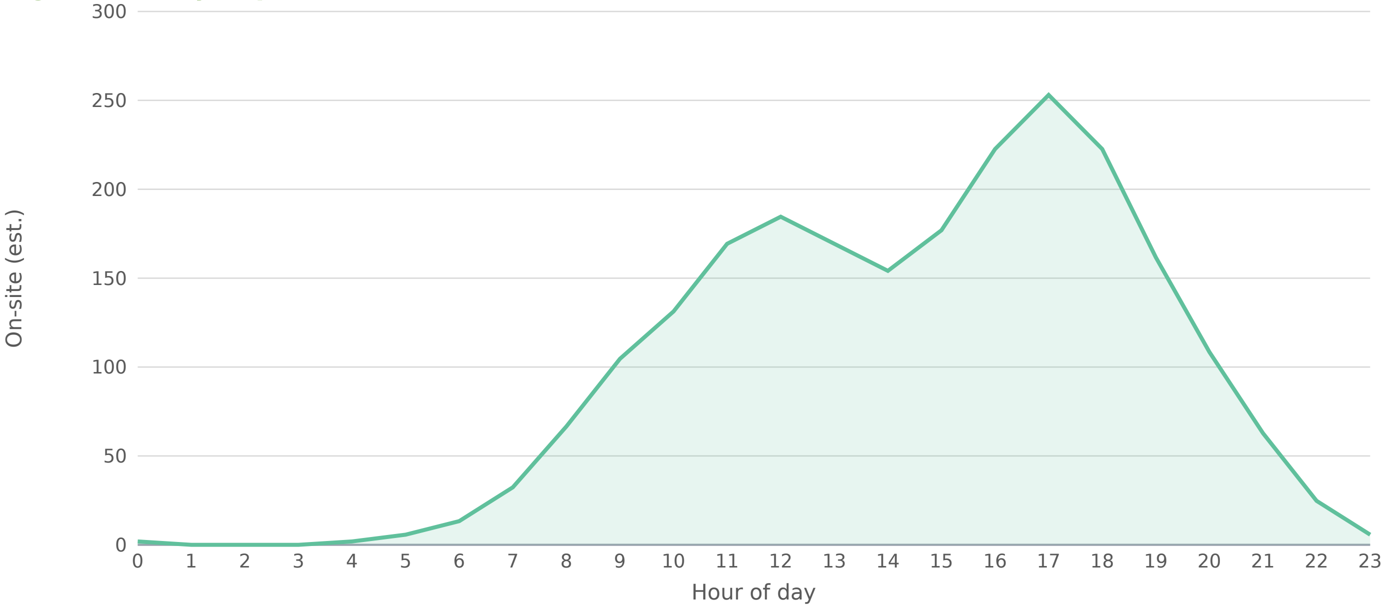


Figure — Daily Trip Accumulation



Peak on-site accumulation ~253 at 17:00, from 6,800 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

4.1 Trip Distribution — Gravity Model

The trip distribution uses the gravity model (term = M / (d^1 · d_site); mass basis: intersection through-volume (AADT × K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

Directional sector	Share of project trips
Northeast (NNE + ENE)	37%
Southeast (ESE + SSE)	36%
Southwest (SSW + WSW)	18%
Northwest (WNW + NNW)	8%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Ash Road & West Higgins Road	NNE	0.88	3,501	0.1	10.99%
Near North Roselle Road	NNE	0.98	3,222	0.1	9.85%
North Roselle Road & Bethel Lane	ESE	0.25	2,457	0.1	9.64%
South Roselle Road & West Schaumburg Road	SSE	0.45	2,457	0.1	8.96%
West Schaumburg Road & North Pleasant Drive	SSE	0.43	2,313	0.1	8.52%
South Roselle Road & Illinois Avenue	SSE	0.65	2,493	0.1	8.51%
North Roselle Road & Bode Road	NNE	0.68	2,457	0.1	8.32%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Branchwood Drive & West Schaumburg Road	SSW	0.50	2,313	0.1	8.31%
North Roselle Road & West Higgins Road	NNE	0.82	2,538	0.1	8.20%
Hilltop Drive & West Schaumburg Road	SSW	0.60	2,313	0.1	8.04%
Grand Canyon Parkway & West Higgins Road	NNW	0.88	2,538	0.1	8.03%
South Salem Drive & West Schaumburg Road	WSW	0.96	594	0.0	1.84%
Summit Drive & North Summit Drive	ESE	0.81	243	0.0	0.79%

13 study-area zone(s). Screening-grade distribution; not a calibrated regional model.

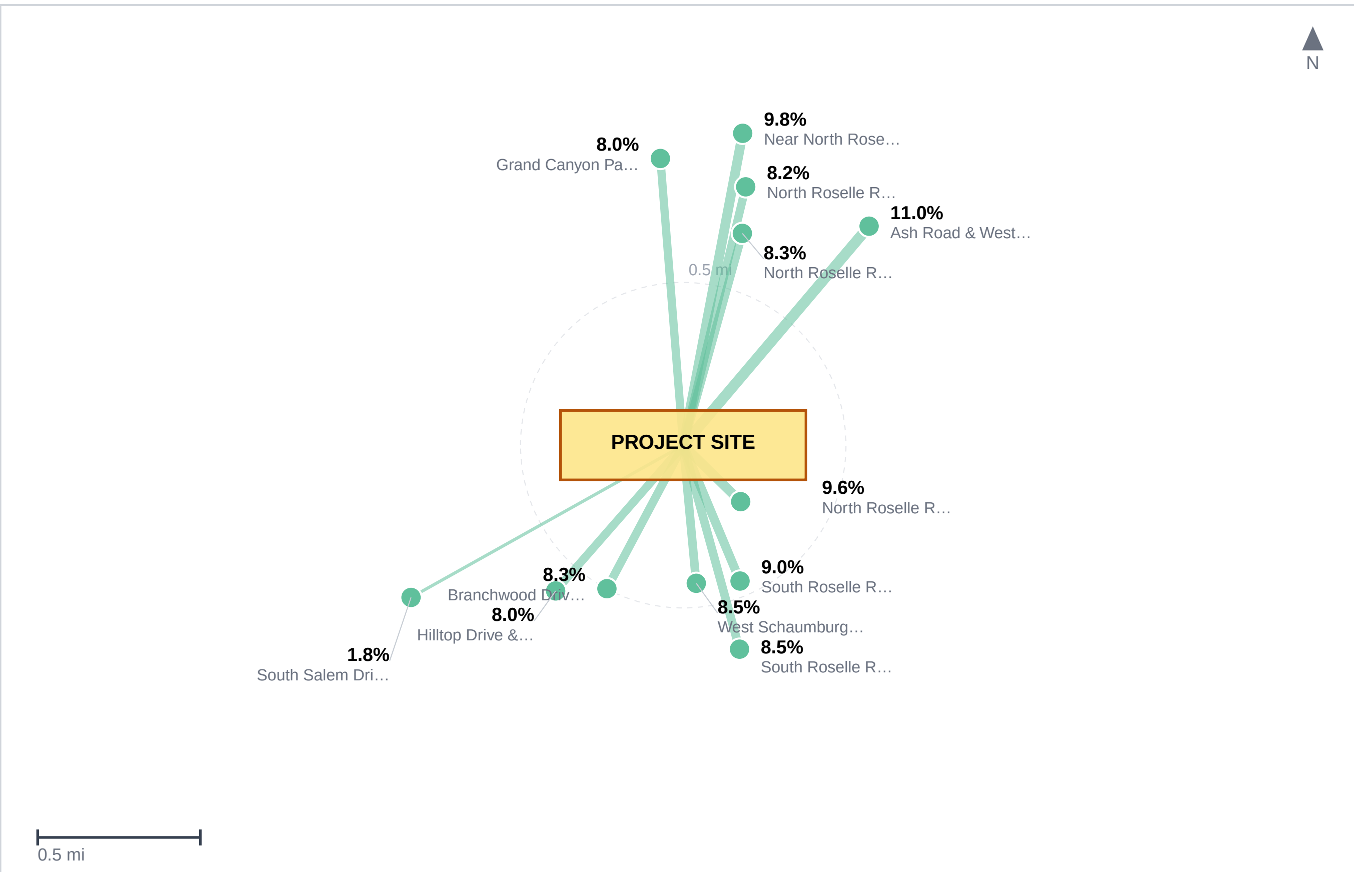
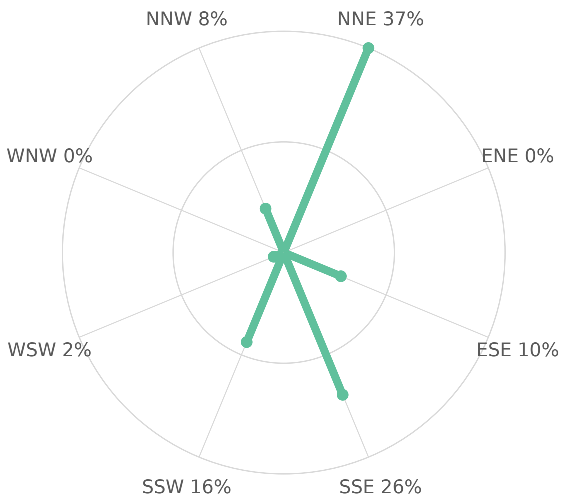


Figure — Project Trip Distribution. Study-area zones plotted to scale at their true bearing and distance from the site; leg weight is proportional to each zone's share of project trips, and the label gives that share. Derived from the Gravity Model distribution — the same shares tabulated above. Screening-grade: zone positions are the analysis locations, not a surveyed base map.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length ∝ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

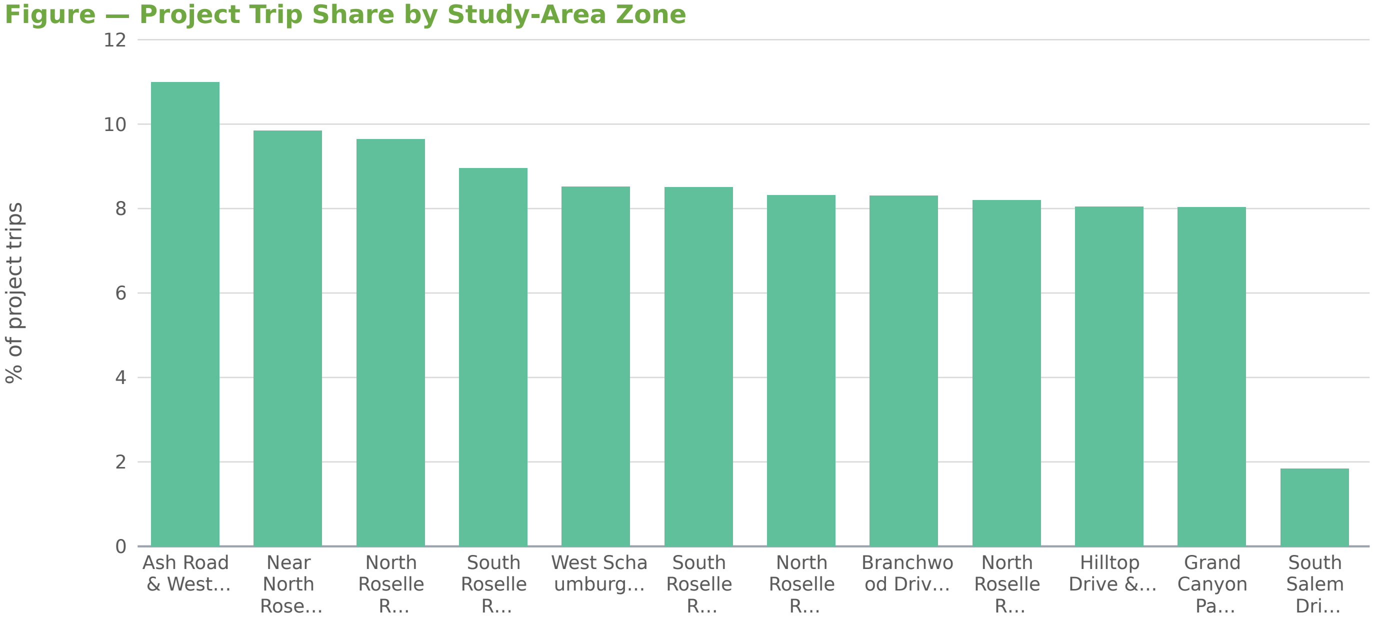
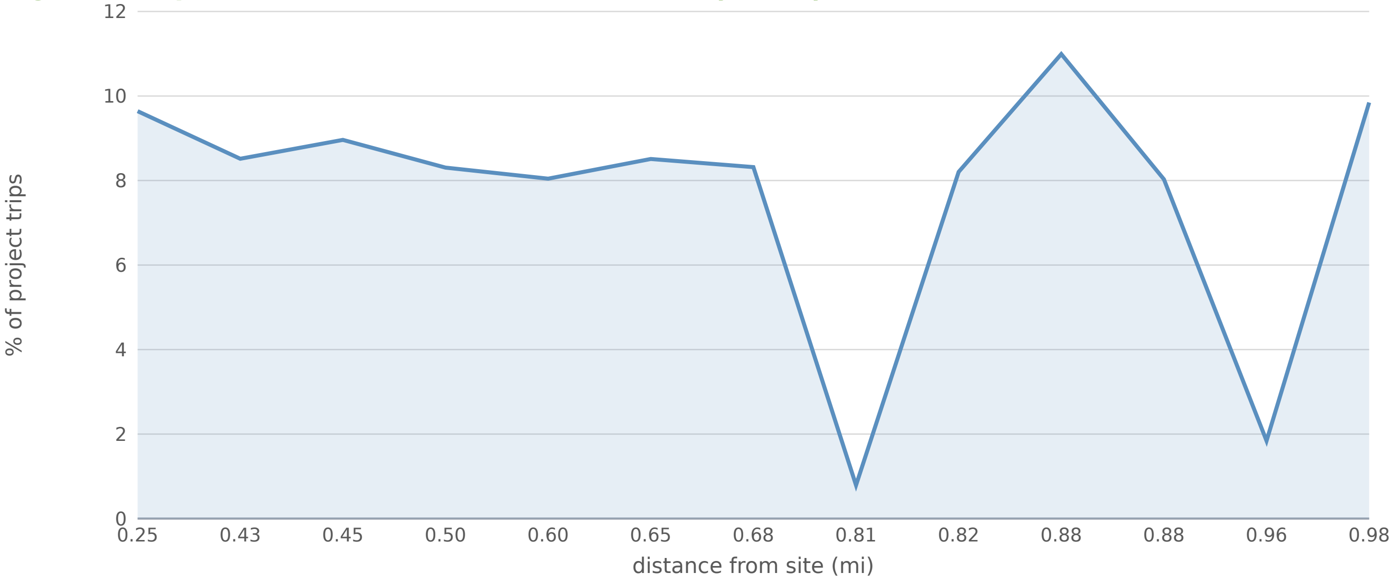


Figure — Trip Share vs. Distance from Site (Gravity Decay)



4.2 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
North Roselle Road & Bethel Lane	ESE	0.25	62	115	29.9%
West Schaumburg Road & North Pleasant Drive	SSE	0.43	5	12	3.1%
South Roselle Road & West Schaumburg Road	SSE	0.45	21	54	14.0%
Branchwood Drive & West Schaumburg Road	SSW	0.50	5	12	3.1%
Hilltop Drive & West Schaumburg Road	SSW	0.60	5	12	3.1%
South Roselle Road & Illinois Avenue	SSE	0.65	16	41	10.6%
North Roselle Road & Bode Road	NNE	0.68	34	51	13.2%
Summit Drive & North Summit Drive	ESE	0.81	8	12	3.1%
North Roselle Road & West Higgins Road	NNE	0.82	22	55	14.3%
Ash Road & West Higgins Road	NNE	0.88	1	3	0.8%
Grand Canyon Parkway & West Higgins Road	NNW	0.88	6	9	2.3%
South Salem Drive & West Schaumburg Road	WSW	0.96	3	7	1.8%
Near North Roselle Road	NNE	0.98	1	2	0.5%

Project trips assigned to each study intersection from the distribution shares above: a four-step gravity model sets the share reaching each intersection from its attraction mass and its distance from the site. Directional load multipliers are a Florida (Caltran) refinement and are not applied in this region — each intersection's loading is taken directly from the gravity weights above.

5.0 BACKGROUND GROWTH

Background traffic is grown at 1.80% per year, derived from the 5-year compound AADT growth rate measured across 646 matched IDOT count stations within the Chicago-Naperville-Elgin MSA bounding box. Source: IDOT Historical AADT, 2020 and 2025 layer snapshots, segments matched by INVENTORY where the AADT vintage year is within ±1 of each layer year. Per-station CAGR distribution: P25 -1.69%/yr · median 1.80%/yr · P75 5.92%/yr. The wide IQR reflects real corridor heterogeneity (heavily developed infill arterials grow faster than legacy state highways in declining counties); for the formal submittal, IDOT D8 Appx. A asks for the per-segment trend on the affected facilities specifically, not the metro median. This screening value is published here for transparency and to give the District a starting point at the kickoff meeting.

6.0 FUTURE CONDITIONS ANALYSIS

Per IDOT District 8 Appendix A, four mandatory scenarios are evaluated for each phase: (1) Opening (Construction) Year No-Build (2027); (2) Opening Year Build (2027); (3) 20-Year Design Year No-Build (2047); (4) 20-Year Design Year Build (2047). For phased developments, a Full-Build-Out year between opening and design year is added. The design year is measured from construction completion, not submittal year, per BLRS §27-6.02(a). This screening assigns Level of Service from the conventional US control-delay bands; a submittal-grade analysis calculates LOS with the District's required methodology and current-edition tools.

Host-jurisdiction LOS standard: BLRS Ch. 32: LOS C controlling for arterials/collectors, LOS D allowed in heavily-developed metro sections, LOS D minimum for urban local streets.

All four D8-mandated scenarios are reported below at each affected intersection. The Design-Year columns project the No-Build and Build conditions forward to 2047 (opening + 20 yr at the same compound growth rate); the project's external trip generation does not grow with the design horizon — only the background traffic does.

Intersection	Existing	Opening NB	Opening Bld	Design NB	Design Bld	Δ delay (s)
North Roselle Road & Bethel Lane	F	F	F	F	F	28.0
West Schaumburg Road & North Pleasant Drive	F	F	F	F	F	2.9
South Roselle Road & West Schaumburg Road	F	F	F	F	F	13.1
Branchwood Drive & West Schaumburg Road	F	F	F	F	F	2.9
Hilltop Drive & West Schaumburg Road	F	F	F	F	F	2.9
South Roselle Road & Illinois Avenue	F	F	F	F	F	10.1
North Roselle Road & Bode Road	F	F	F	F	F	12.3
Summit Drive & North Summit Drive	B	B	B	B	B	0.0
North Roselle Road & West Higgins Road	F	F	F	F	F	13.4
Ash Road & West Higgins Road	F	F	F	F	F	0.0
Grand Canyon Parkway & West Higgins Road	F	F	F	F	F	2.2
South Salem Drive & West Schaumburg Road	B	B	B	B	B	0.1

Intersection	Existing	Opening NB	Opening Bld	Design NB	Design Bld	Δ delay (s)
Near North Roselle Road	F	F	F	F	F	0.0

7.0 MITIGATION, WARRANTS, AND SIGHT DISTANCE

Recommended improvements below are screening-level concepts sized to the projected delay change. The formal D8-style submittal additionally requires explicit turn-lane warrant analysis (BDE nomographs), ILMUTCD signal warrant analysis (Warrants 1–9 with met/not-met), sight-distance verification (BLRS Ch. 28: SSD, ISD), and auxiliary-lane / acceleration / deceleration geometry per BLRS Ch. 34. Pedestrian and bicycle accommodations are evaluated against BDE Ch. 17 non-motorized warrants.

North Roselle Road & Bethel Lane [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

West Schaumburg Road & North Pleasant Drive [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

South Roselle Road & West Schaumburg Road [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Branchwood Drive & West Schaumburg Road [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Hilltop Drive & West Schaumburg Road [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

South Roselle Road & Illinois Avenue [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

North Roselle Road & Bode Road [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

North Roselle Road & West Higgins Road [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Ash Road & West Higgins Road [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Grand Canyon Parkway & West Higgins Road [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Near North Roselle Road [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

8.0 PROGRAMMED PROJECTS

Review of programmed transportation projects within the study area should consult: IDOT MYP FY 2026–2031, CMAP TIP, and Cook County DOTH project list. This screening analysis does not automatically integrate programmed-projects data; manual review against the IDOT MYP GIS layer (gis-idot.opendata.arcgis.com) is recommended for any submittal.

9.0 CONCLUSIONS & RECOMMENDATIONS

This screening TIS identifies 0 intersections with LOS drops and 11 operating at LOS E or F under build conditions. The formal submittal to Cook County DOTH should validate these screening results against current-edition manuals, fresh TMCs within the most recent 12 months, derived growth rates, the four-scenario horizon analysis, and the host-jurisdiction scoping outcome. The report must be sealed by a Licensed Professional Engineer of Illinois; digital seals and signatures are allowed per 68 Ill. Admin. Code §1380.295. The required submittal package is two bound paper copies + one electronic PDF + the electronic capacity-analysis source files (Synchro / HCS / Vistro), routed to the District Permits Unit Chief. Allow approximately 8–10 weeks per submittal review and 18–24 months total for signalized or widening projects.

FINDINGS

• Project will generate 6,800 new daily vehicle trips, with 680 during the PM peak hour (326 inbound / 354 outbound).

• Pass-by credit 30% and internal-capture credit 0% applied at the PM peak (25% of that credit at the AM and Saturday-midday periods, per the industry rule of thumb that off-peak shopping diverts less) before off-site assignment (standard pass-by methodology; ULI Internal Capture).

• Auto-mode share of 49% applied to external trips for Chicago-Naperville-Elgin MSA; 51% of generated trips are assumed to arrive by transit, walking, or cycling and do not load the off-site roadway network. Source: ACS 5-Year Estimates Table B08301 (US) or national-stats equivalent.

• Existing volumes were grown by 1.50%/yr over 1 year (×1.01) to the opening-year horizon.

• 13 signalized intersections fall within the study area; 0 are projected to drop at least one LOS grade after build-out.

• 11 intersections are projected to operate at LOS E or F under the build condition and require formal mitigation per Chicago Department of Transportation (CDOT) TIS guidance.

• Worst-impact location: North Roselle Road & Bethel Lane — projected delay rises 28.0s (LOS F → F); 95th-pct queue 725 ft on the critical approach.

• Conclusions are reported at the applied screening assumptions. Discrete-scenario sensitivity at the formal TIA scoping meeting should test: trip-generation method (rate vs. fitted-curve equation); internal capture and pass-by credit variants; and a ±0.5%/yr background growth band around the applied value.

METHODOLOGY NOTES

• Trip generation uses public-data average rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) for the selected land-use code, computed for AM peak, PM peak, Saturday midday, and daily totals. Saturday-midday rates are estimated as a published industry multiple of the PM peak rate by land-use category.

• Pass-by and internal-capture credits are applied in full at the PM peak (and at 25% of that credit fraction for the AM and Saturday-midday periods) per standard pass-by screening methodology and ULI Mixed-Use Internal Capture defaults; only the residual external vehicle trips are assigned to off-site intersections.

• Existing intersection volumes are grown to the opening-year horizon at the user-supplied annual growth rate (default 1.5%/yr) before the capacity analysis.

• Weather adjustment applies published rain/snow capacity-reduction factors as adopted in US traffic-engineering practice: clear 1.00, light rain 0.95, heavy rain 0.86, light snow 0.86, heavy snow 0.70. The factor multiplies the saturation flow at every intersection.

• Off-site impact is screened for all signalized intersections within the study radius (default 0.5 mi) using the four-step travel demand model (FHWA; NCHRP Report 716). Step 1 Trip Generation: public-data average rates (SANDAG 2002 / NHTS 2017 / NCHRP 716) give the site's external (post pass-by / internal-capture) productions. Step 2 Trip Distribution: a production-constrained gravity model $T_j = P \cdot (A_j \cdot F_j) / \Sigma(A_x \cdot F_x)$ allocates trips to surrounding signals, where attractiveness A_j is the signal's through-volume and the friction factor F_j is the NCHRP-716 gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ (home-based-work coefficients a=28507, b=-0.02, c=-0.123) on the travel time t to each signal. Step 3 Mode Choice: a binary logit $P(auto)=1/(1+e^{-(ASC-\lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (a density proxy from surrounding through-volumes) so denser, more transit-served sites split further from auto; only the resulting vehicle trips load the network. Step 4 Route Assignment: a capacity-constrained assignment using the BPR volume-delay function $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives. Signals lacking AADT data fall back to a constant 5,000 vpd attraction.

• After assignment, all signalized intersections within the study radius are reported in the affected-intersections table. Project-added trips, v/c ratio change, control delay change, LOS change, and 95th-percentile queue are reported for each intersection so the reviewer can assess relative impact. Intersections beyond the study radius are excluded from analysis.

• Auto-mode share is applied per metro before assignment. Suburban-US metros default to 90% auto (ACS 5-Year B08301 median); transit-heavy metros use measured auto-mode share (e.g., NYC 32%, Tokyo 30%, London 38%, San Francisco 47%). Non-auto trips (transit, walking, cycling) do not load the off-site roadway. This is a screening-level adjustment; a real TIS submittal in a transit-heavy market should refine with project-specific TAZ data.

• Candidate signals are de-duplicated within a 45m clustering threshold to prevent OSM divided-arterial splits and way-record artifacts from double-counting a single physical intersection.

• Intersection-level control delay uses the standard signalized-intersection model d = d1 + d2 (Webster uniform delay + Akçelik time-dependent incremental-delay term, both openly published) with a 90s cycle, g/C = 0.45, 1,800 vphpl saturation flow (× weather factor), 15-minute peak analysis period (T = 0.25 hr) and pretimed-signal incremental-delay factor k = 0.5. Reported control delay is capped at 300 s (LOS F) for screening reliability — the incremental-delay term is not calibrated far above capacity, so oversaturated approaches are reported as LOS F rather than an implausible delay figure; a calibrated design-level analysis (HCS / Synchro) supersedes this screen.

• Approach-level analysis splits each signal's inflow across NB/SB/EB/WB approaches (deterministic per-signal allocation perturbed ±15% from a 30/25/25/20 base) and assigns added trips to each approach by cosine-similarity to the bearing of the project relative to the signal. Per-approach v/c, control delay, LOS, and 95th-percentile back-of-queue length (standard cyclic-queue relation with Poisson percentile factor: Q95 ≈ Q1 × 1.65 × 25 ft/veh) are reported.

• Level of Service is assigned from the conventional US signalized control-delay bands republished in state DOT traffic-engineering manuals: A ≤10s, B ≤20s, C ≤35s, D ≤55s, E ≤80s, F >80s.

• Sensitivity is reported in narrative form per standard TIA practice: trip-generation method (rate vs. fitted-curve equation), discrete internal capture and pass-by credit variants, and a ±0.5%/yr growth-rate band around the applied value. The engine retains an internal stochastic-sensitivity routine (Box-Muller-perturbed trip rate and existing volume) for demo-mode diagnostics; it is not surfaced in the deliverable because TIA sensitivity is conducted through discrete scoping-agreed scenario variants, not statistical perturbation of unmeasured distributions.

• Mitigations are screening-level recommendations sized to the projected delay change, not full Synchro/SimTraffic optimization runs. A formal TIS submittal should validate these recommendations with detailed traffic counts and signal-timing analysis.

• Turbo-lane (continuous-green-T) screening flags signalized 3-leg T-intersections where one or more main-street through lanes could flow continuously while the minor-street left turn merges in the median — recovering the green time the through movement would otherwise lose to the signal. Candidacy requires measured 3-leg geometry, an arterial-class main street, and a detected median (divided carriageway), all derived from the OpenStreetMap road network. Approach-capacity gain is computed as (turbo lanes / approach lanes) × (1 − g/C) / (g/C), with the main-street through g/C derived from the modeled main-vs-minor critical-flow split; the result is reported within the +7%…+173% envelope documented in the continuous-green-T design literature — the standard national references for this geometry (David Plummer & Associates, Adding Turbo Lanes to T-Intersections, 2010; and the Design Guidelines for the Development of Continuous Green Intersections, 1997). Lane counts and signal timing must be field-verified before design.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: T_j = P · (A_j · F_j) / Σ (A_x · F_x). Attractiveness A_j is each signal's through-volume; the friction factor F_j is a gamma function F = a·t^b·e^(c·t) — the form recommended by NCHRP Report 716 — on the travel time t to the signal, with home-based-work shape parameters b=-0.02, c=-0.123 (consistent with NCHRP 716's sampled MPO gravity parameters; the scale constant a cancels in the share normalization).

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
North Roselle Road & Bethel Lane	0.25	0.6	115	29.9%
North Roselle Road & West Higgins Road	0.82	2.0	55	14.3%
South Roselle Road & West Schaumburg Road	0.45	1.1	54	14.0%
North Roselle Road & Bode Road	0.68	1.6	51	13.2%
South Roselle Road & Illinois Avenue	0.65	1.6	41	10.6%
West Schaumburg Road & North Pleasant Drive	0.43	1.0	12	3.1%
Branchwood Drive & West Schaumburg Road	0.50	1.2	12	3.1%
Hilltop Drive & West Schaumburg Road	0.60	1.4	12	3.1%
Summit Drive & North Summit Drive	0.81	1.9	12	3.1%
Grand Canyon Parkway & West Higgins Road	0.88	2.1	9	2.3%
South Salem Drive & West Schaumburg Road	0.96	2.3	7	1.8%
Ash Road & West Higgins Road	0.88	2.1	3	0.8%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 49% applied via a binary logit P(auto) = 1 / (1 + e^-(ASC − λ·ΔGC)) calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 51% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

Project trips are routed over the local road network by shortest path and loaded with a capacity-constrained BPR equilibrium (volume-delay t = t0·[1 + 0.15·(v/c)^4], Method of Successive Averages), so congested links shift trips toward alternative routes. 100% of study signals lie on a modeled route; highest loaded-link v/c: 0.22. Corridors carrying the project trips:

Corridor (road class)	Loaded length mi	PM project vph	v/c
Collector	0.82	127	0.22
Minor arterial	3.04	106	0.21
Principal arterial	0.68	26	0.16

Network shortest-path assignment over the OSM road graph within the study area; per-signal v/c, delay, LOS, and queue are in the capacity appendix.

Conserved assignment: project trips routed to 15 cordon gateways on the study boundary (weighted by the directional distribution above). Flow conservation verified at render input: Σ entering = Σ leaving at 217 network junctions, max imbalance 0.00 veh. 11 study intersection(s) carry path-derived movements; 2 retain the geometric octant model (labeled per intersection).

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach capacity table. Analysis uses the openly-published Webster/Akcelik signalized control-delay method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

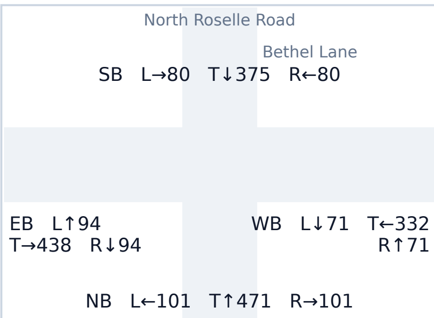
Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 North Roselle Road & Bethel Lane

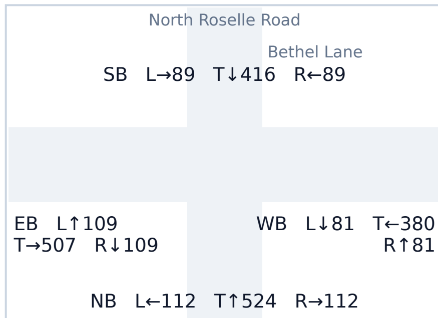
Signal ID	chicago-3303
Location	NW Chicago-Naperville-Elgin · 0.25 mi from site
Added PM peak trips	115
Existing / No-Build (PM)	LOS F · 205.7 s/veh · v/c 1.39
Build (PM)	LOS F · 233.7 s/veh · v/c 1.45
Δ control delay	+28.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

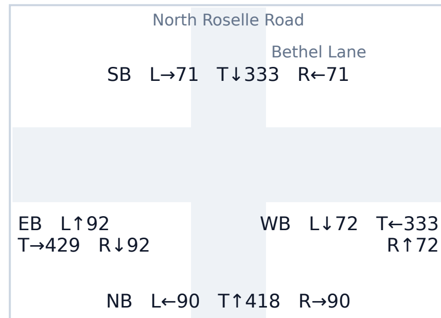
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	748	0	748	0.92 → 0.92	40.9 → 40.9	D → D	725
SB	594	0	594	0.73 → 0.73	26.1 → 26.1	C → C	503
EB	671	54	725	0.83 → 0.90	31.3 → 37.3	C → D	689
WB	481	61	542	0.59 → 0.67	21.8 → 23.8	C → C	440

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	61	53%
EB Right	54	47%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

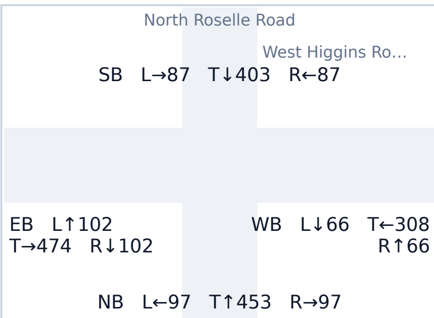
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 North Roselle Road & West Higgins Road

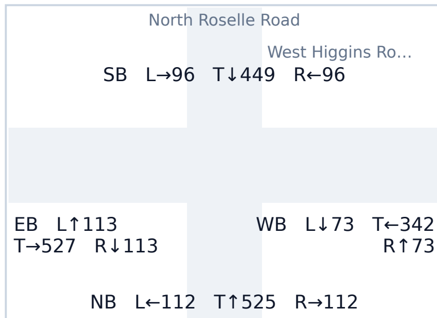
Signal ID	chicago-9796
Location	NW Chicago-Naperville-Elgin · 0.82 mi from site
Added PM peak trips	55
Existing / No-Build (PM)	LOS F · 225.7 s/veh · v/c 1.43
Build (PM)	LOS F · 239.1 s/veh · v/c 1.46
Δ control delay	+13.4 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

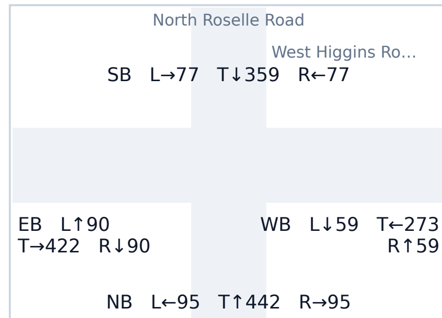
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	694	55	749	0.86 → 0.92	33.5 → 41.2	C → D	728
SB	641	0	641	0.79 → 0.79	28.9 → 28.9	C → C	564
EB	753	0	753	0.93 → 0.93	41.9 → 41.9	D → D	734
WB	488	0	488	0.60 → 0.60	22.0 → 22.0	C → C	380

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	55	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

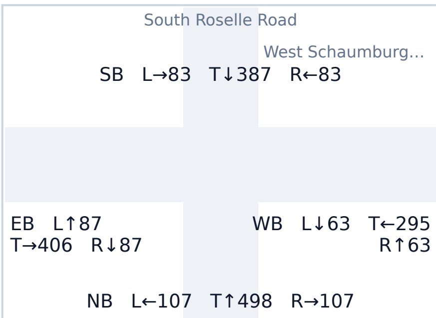
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 South Roselle Road & West Schaumburg Road

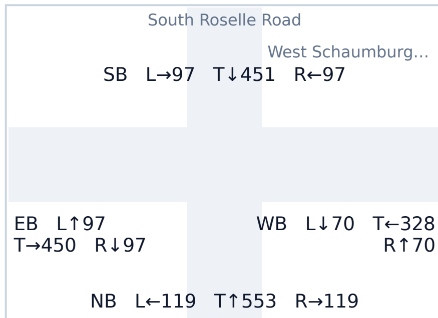
Signal ID	chicago-9798
Location	NW Chicago-Naperville-Elgin · 0.45 mi from site
Added PM peak trips	54
Existing / No-Build (PM)	LOS F · 205.7 s/veh · v/c 1.39
Build (PM)	LOS F · 218.8 s/veh · v/c 1.42
Δ control delay	+13.1 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

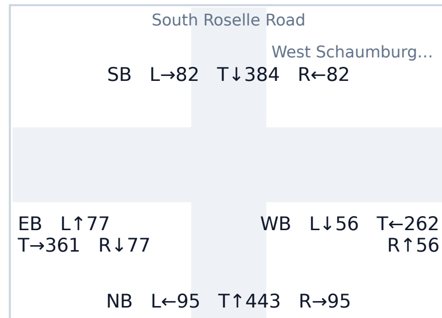
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	791	0	791	0.98 → 0.98	50.8 → 50.8	D → D	801
SB	591	54	645	0.73 → 0.80	26.0 → 29.2	C → C	570
EB	644	0	644	0.80 → 0.80	29.1 → 29.1	C → C	569
WB	468	0	468	0.58 → 0.58	21.4 → 21.4	C → C	358

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	54	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

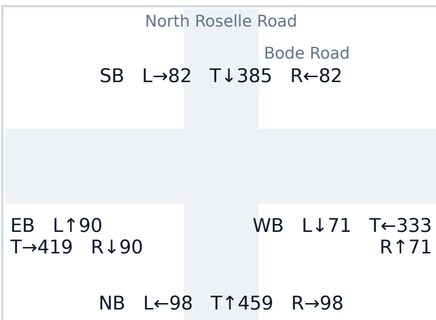
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 North Roselle Road & Bode Road

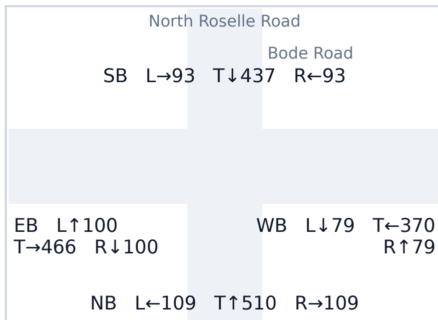
Signal ID	chicago-3295
Location	NW Chicago-Naperville-Elgin · 0.68 mi from site
Added PM peak trips	51
Existing / No-Build (PM)	LOS F · 205.7 s/veh · v/c 1.39
Build (PM)	LOS F · 218.0 s/veh · v/c 1.41
Δ control delay	+12.3 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

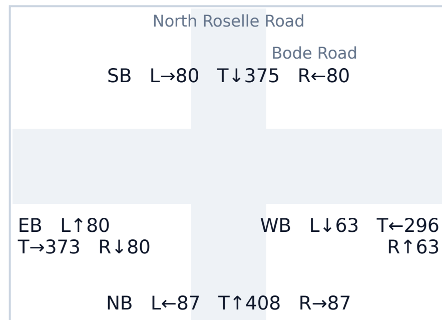
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	728	0	728	0.90 → 0.90	37.7 → 37.7	D → D	693
SB	573	51	623	0.71 → 0.77	25.1 → 27.8	C → C	541
EB	666	0	666	0.82 → 0.82	30.8 → 30.8	C → C	600
WB	528	0	528	0.65 → 0.65	23.3 → 23.3	C → C	424

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	51	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

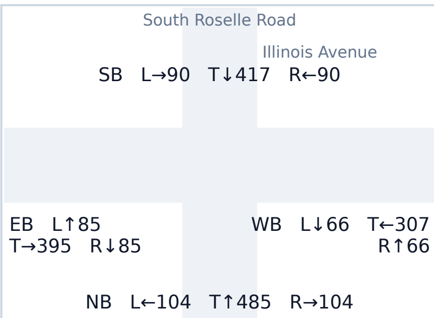
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 South Roselle Road & Illinois Avenue

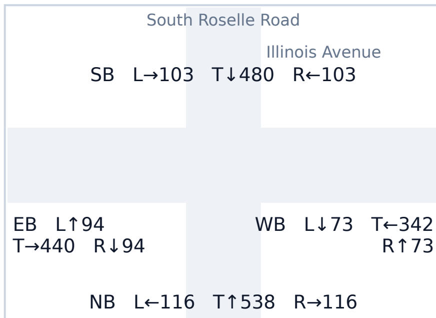
Signal ID	chicago-2345
Location	NW Chicago-Naperville-Elgin · 0.65 mi from site
Added PM peak trips	41
Existing / No-Build (PM)	LOS F · 214.5 s/veh · v/c 1.41
Build (PM)	LOS F · 224.6 s/veh · v/c 1.43
Δ control delay	+10.1 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

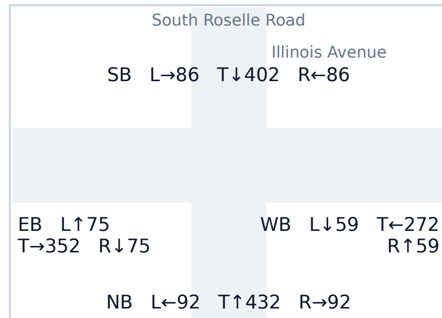
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	770	0	770	0.95 → 0.95	45.5 → 45.5	D → D	763
SB	645	41	686	0.80 → 0.85	29.2 → 32.6	C → C	629
EB	628	0	628	0.78 → 0.78	28.1 → 28.1	C → C	547
WB	488	0	488	0.60 → 0.60	22.0 → 22.0	C → C	379

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	41	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

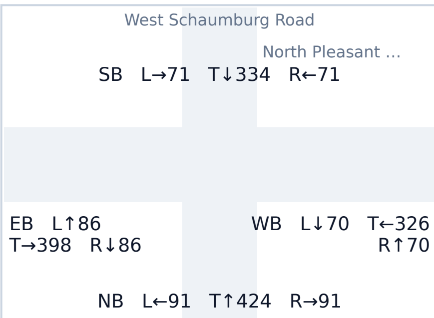
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 West Schaumburg Road & North Pleasant Drive

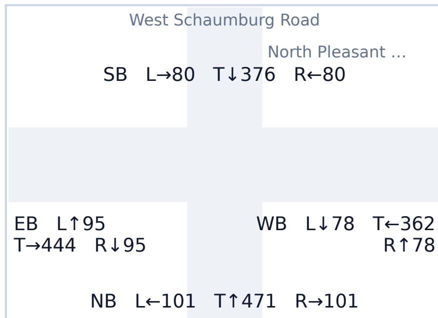
Signal ID	chicago-2344
Location	NW Chicago-Naperville-Elgin · 0.43 mi from site
Added PM peak trips	12
Existing / No-Build (PM)	LOS F · 170.4 s/veh · v/c 1.30
Build (PM)	LOS F · 173.3 s/veh · v/c 1.31
Δ control delay	+2.9 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

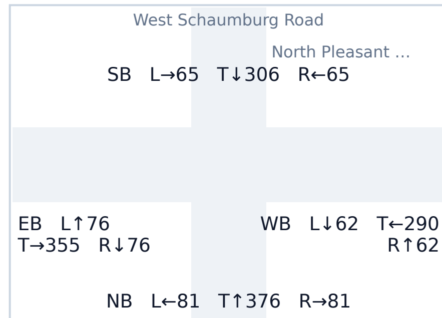
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	673	0	673	0.83 → 0.83	31.4 → 31.4	C → C	610
SB	523	12	536	0.65 → 0.66	23.2 → 23.6	C → C	432
EB	634	0	634	0.78 → 0.78	28.4 → 28.4	C → C	555
WB	518	0	518	0.64 → 0.64	23.0 → 23.0	C → C	412

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Right	12	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

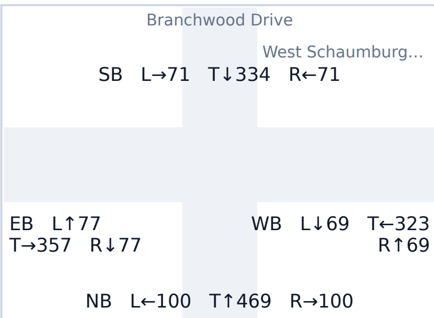
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.7 Branchwood Drive & West Schaumburg Road

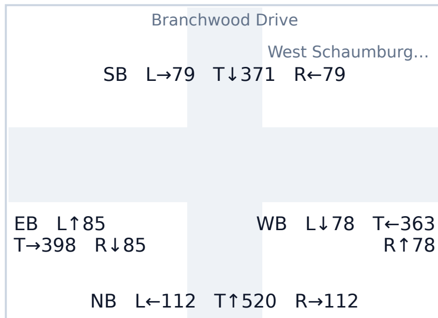
Signal ID	chicago-2292
Location	NW Chicago-Naperville-Elgin · 0.50 mi from site
Added PM peak trips	12
Existing / No-Build (PM)	LOS F · 170.4 s/veh · v/c 1.30
Build (PM)	LOS F · 173.3 s/veh · v/c 1.31
Δ control delay	+2.9 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

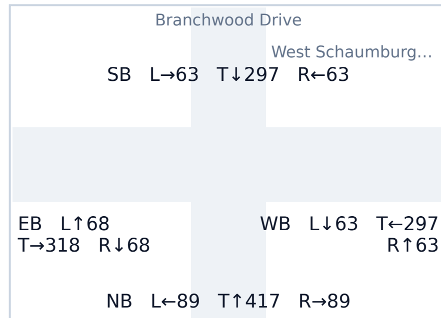
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	744	0	744	0.92 → 0.92	40.2 → 40.2	D → D	719
SB	529	0	529	0.65 → 0.65	23.4 → 23.4	C → C	425
EB	568	0	568	0.70 → 0.70	24.9 → 24.9	C → C	470
WB	507	12	519	0.63 → 0.64	22.6 → 23.0	C → C	414

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	12	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

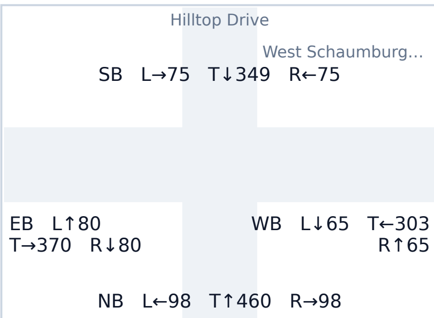
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.8 Hilltop Drive & West Schaumburg Road

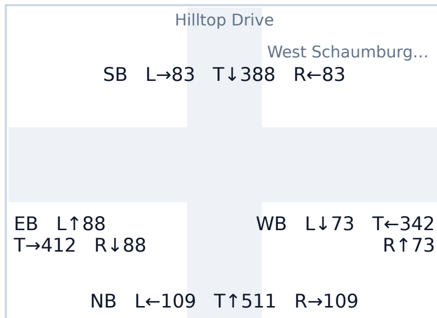
Signal ID	chicago-2304
Location	NW Chicago-Naperville-Elgin · 0.60 mi from site
Added PM peak trips	12
Existing / No-Build (PM)	LOS F · 170.4 s/veh · v/c 1.30
Build (PM)	LOS F · 173.3 s/veh · v/c 1.31
Δ control delay	+2.9 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

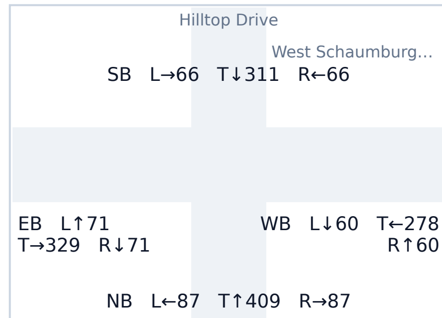
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	729	0	729	0.90 → 0.90	37.9 → 37.9	D → D	695
SB	554	0	554	0.68 → 0.68	24.3 → 24.3	C → C	454
EB	588	0	588	0.73 → 0.73	25.9 → 25.9	C → C	496
WB	476	12	488	0.59 → 0.60	21.6 → 22.0	C → C	380

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	12	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

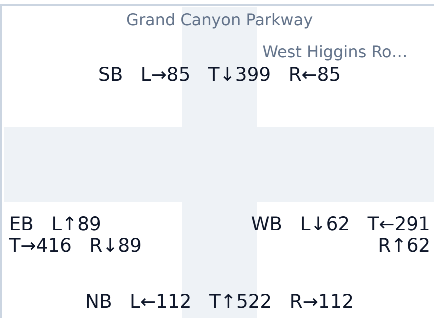
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.9 Grand Canyon Parkway & West Higgins Road

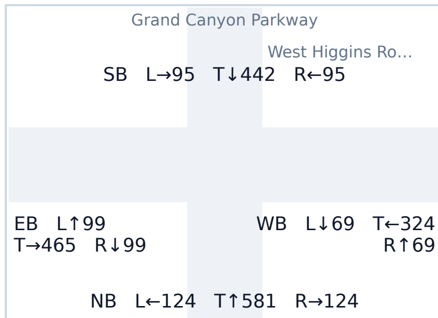
Signal ID	chicago-4662
Location	NW Chicago-Naperville-Elgin · 0.88 mi from site
Added PM peak trips	9
Existing / No-Build (PM)	LOS F · 225.7 s/veh · v/c 1.43
Build (PM)	LOS F · 227.9 s/veh · v/c 1.44
Δ control delay	+2.2 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

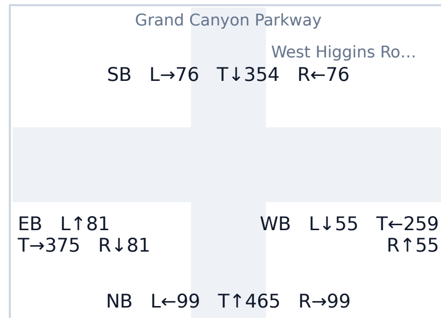
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	829	0	829	1.02 → 1.02	62.1 → 62.1	E → E	847
SB	632	0	632	0.78 → 0.78	28.4 → 28.4	C → C	553
EB	654	9	663	0.81 → 0.82	29.8 → 30.5	C → C	595
WB	462	0	462	0.57 → 0.57	21.2 → 21.2	C → C	352

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	9	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

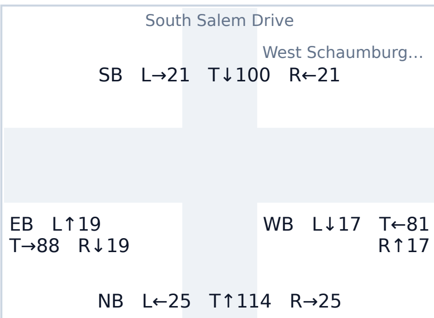
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.10 South Salem Drive & West Schaumburg Road

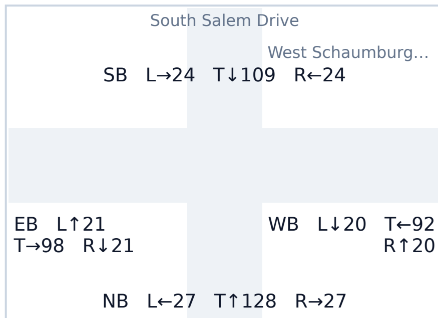
Signal ID	chicago-4393
Location	NW Chicago-Naperville-Elgin · 0.96 mi from site
Added PM peak trips	7
Existing / No-Build (PM)	LOS B · 17.1 s/veh · v/c 0.33
Build (PM)	LOS B · 17.2 s/veh · v/c 0.34
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

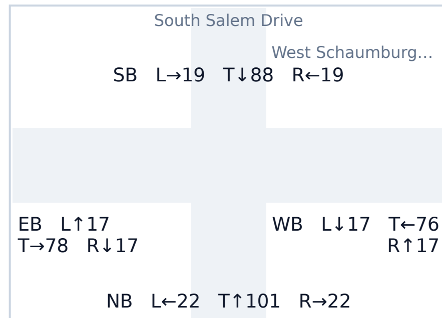
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	182	0	182	0.22 → 0.22	15.8 → 15.8	B → B	115
SB	157	0	157	0.19 → 0.19	15.5 → 15.5	B → B	98
EB	140	0	140	0.17 → 0.17	15.2 → 15.2	B → B	86
WB	124	7	132	0.15 → 0.16	15.0 → 15.1	B → B	80

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Left	4	57%
WB Through	3	43%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

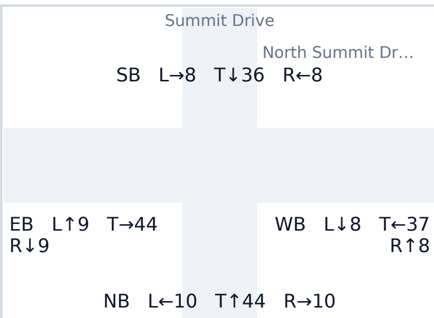
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.11 Summit Drive & North Summit Drive

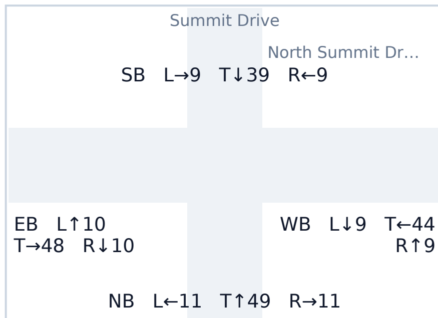
Signal ID	chicago-7380
Location	NW Chicago-Naperville-Elgin · 0.81 mi from site
Added PM peak trips	12
Existing / No-Build (PM)	LOS B · 14.9 s/veh · v/c 0.14
Build (PM)	LOS B · 14.9 s/veh · v/c 0.14
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

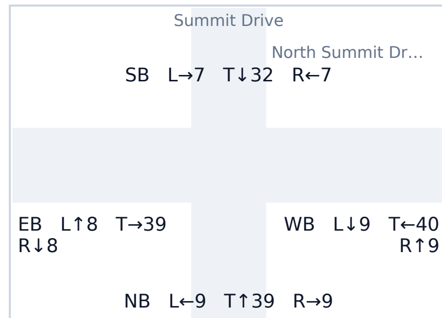
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	71	0	71	0.09 → 0.09	14.4 → 14.4	B → B	42
SB	57	0	57	0.07 → 0.07	14.2 → 14.2	B → B	34
EB	68	0	68	0.08 → 0.08	14.4 → 14.4	B → B	40
WB	50	12	62	0.06 → 0.08	14.2 → 14.3	B → B	36

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	12	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

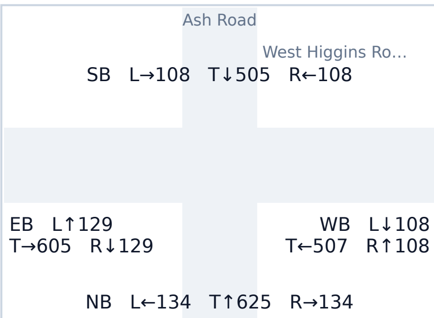
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.12 Ash Road & West Higgins Road

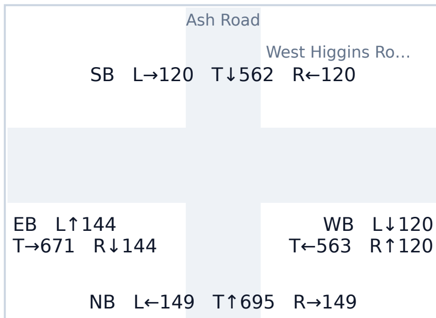
Signal ID	chicago-4870
Location	NW Chicago-Naperville-Elgin · 0.88 mi from site
Added PM peak trips	3
Existing / No-Build (PM)	LOS F · 300.0 s/veh · v/c 1.97
Build (PM)	LOS F · 300.0 s/veh · v/c 1.98
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

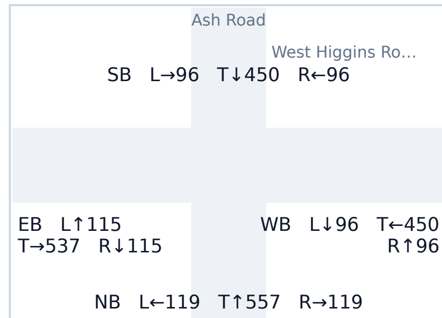
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	991	2	993	1.22 → 1.23	136.2 → 137.0	F → F	1,016
SB	801	1	802	0.99 → 0.99	53.5 → 53.9	D → D	820
EB	959	0	959	1.18 → 1.18	119.7 → 119.7	F → F	981
WB	803	0	803	0.99 → 0.99	54.1 → 54.1	D → D	821

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	2	67%
SB Through	1	33%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

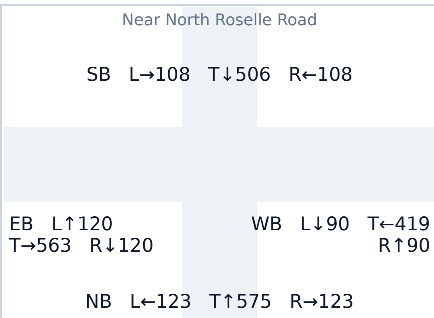
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.13 Near North Roselle Road

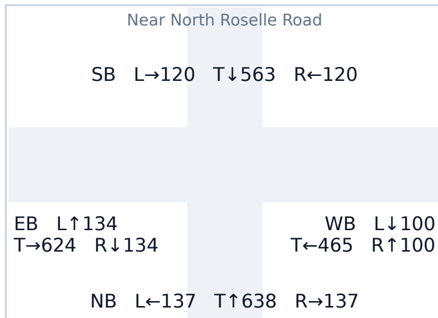
Signal ID	chicago-8866
Location	NW Chicago-Naperville-Elgin · 0.98 mi from site
Added PM peak trips	2
Existing / No-Build (PM)	LOS F · 300.0 s/veh · v/c 1.82
Build (PM)	LOS F · 300.0 s/veh · v/c 1.82
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

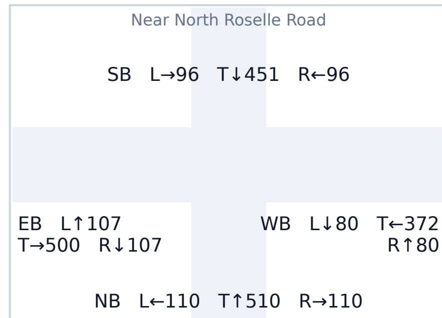
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	911	1	912	1.12 → 1.13	96.4 → 96.9	F → F	933
SB	802	1	803	0.99 → 0.99	53.8 → 54.0	D → D	821
EB	892	0	892	1.10 → 1.10	87.7 → 87.7	F → F	913
WB	665	0	665	0.82 → 0.82	30.7 → 30.7	C → C	598

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	1	50%
SB Through	1	50%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.